

SUMMARY

Backend Engineer with production experience building **real-money trading systems** and applied **GenAI pipelines**. Comfortable across Python and Go, with end-to-end ownership of concurrency-safe order execution, asynchronous/background processing, RAG and agentic LLM systems, domain-driven system design, and infrastructure security (TLS hardening, auth design). Independently designs and ships complete systems — from data modeling to production deployment.

EXPERIENCE

Software Engineer Intern — Potoos (Algorithmic Trading Platform & Research Services)

Sept 2025 – Present

Algorithmic Trading Platform (Django)

- Primary backend contributor (**69% of commits, ~67K lines of authored code**) to a live production **algorithmic trading platform** handling **real user capital**, alongside one other engineer.
- Built and maintained asynchronous pipelines using **Django, Celery, and Redis** to handle order placement, stop-loss, modification, and exit flows.
- Used **Redis and Lua scripts** to guarantee concurrency-safe rule state, eliminating duplicate or conflicting order executions.
- Designed execution logic supporting multiple brokers through a shared rule and dispatch layer.
- Deployed and operated **Dockerized services behind Nginx** on Linux servers.

Research Services — Go Microservice

- Independently designed and built a **Go/Gin microservice (96% of commits, 15.8K+ LOC)** handling KYC verification, risk profiling, and trade-signal distribution for a **SEBI-regulated research advisory product**, running alongside the existing Django backend.
- Automated KYC status verification using **headless Chrome (chromedp)** with anti-detection evasion, async result polling, PDF generation, and **MinIO** storage.
- Built a scored risk-profiling engine gating access to cash-equity vs. F&O segments per **SEBI-style compliance rules**.
- Built a CSV-based multi-leg strategy ingestion pipeline with Google Drive → MinIO asset migration, per-row transactional isolation, and cross-service sync to the Django backend.
- Implemented a WhatsApp broadcast worker with templated messaging and Redis-queued dispatch; hardened OTP auth flow (rate limiting, cooldowns, IP abuse blocking) and JWT session versioning.
- Authored the production **Nginx** layer: **TLS 1.2/1.3 hardening**, strict cipher suites, and Host-header allow-listing preventing header-injection and virtual-host leakage.

PROJECTS

RheumaLink | Healthcare Practice Management Platform | [Django](#), [DRF](#), [Celery](#), [Channels](#), [FastAPI](#), [Docker](#) [Demo](#)

- Architected and built a full-stack clinic management platform for rheumatology practices, independently studying real clinical intake workflows and disease-specific diagnostic criteria (**DAS28, HAQ**) to design a domain-accurate **RUMAT diagnosis model spanning 100+ structured clinical fields**, paired with an LLM-generated clinical summary streamed live from the AI microservice.
- Built a real-time appointment and queue system using **Django Channels/WebSockets**, with automatic per-doctor daily token generation and live dashboard updates.
- Designed an interactive **clickable SVG joint chart** — a human skeleton diagram where each of the 28 **DAS28** joints cycles through tender/swollen states with live color feedback — feeding directly into a **DAS28 disease-activity scoring engine** that computes composite scores from joint counts and lab markers (ESR/CRP).
- Built a real-time **voice-to-prescription pipeline**: streamed doctor dictation over **WebSockets** to a self-hosted **MedASR** model, then used **Groq** LLMs to correct medical pronunciation and auto-structure the note into complaints, findings, diagnosis, and advice fields live on the dashboard.
- Built the prescription delivery pipeline: injected structured data into HTML templates, converted to PDF via a self-hosted **Gutenberg (headless Chromium)** microservice, and auto-dispatched to patients over **WhatsApp** via a Celery-queued task; supports multiple independent clinics, each with its own API key and admin-toggled subscription.
- Built an asynchronous lab-report pipeline (Celery + Redis) integrating **OCR** for structured data extraction from PDF/image reports, and a **WhatsApp Business webhook** for patient intake with session-state management and automated triage responses.

ColdMail AI | RAG-Powered Cold Email Agent | [Python](#), [FastAPI](#), [LangChain](#), [pgvector](#), [Groq](#) [GitHub](#) | [Demo](#)

- Built a self-hosted layout-aware RAG system that parses multiple resume variants (AI/Backend/Fullstack) via **Docling**, generating local embeddings with **HuggingFace all-MiniLM-L6-v2** (384-dim) stored in **PostgreSQL/pgvector** for cosine-similarity retrieval.
- Designed the generation pipeline to embed incoming job descriptions, retrieve the top-8 matching resume chunks, and draft tailored cold emails using **LangChain + Groq (Llama 3.3 70B)**.
- Built an async resume-ingestion pipeline with live status polling and a class-based controller architecture (settings, resume, email), plus a background SMTP SSL mailer (port 465) that compiles Markdown drafts into styled HTML MIME emails with PDF resume attachments for immediate dispatch — all across a Dockerized FastAPI + pgvector stack.

Society Application | CSI (Computer Society of India) Backend, AKGEC | [Django](#), [DRF](#), [WebSockets](#) [GitHub](#) | [Demo](#)

- Built the official modular Django backend for CSI's chapter at AKGEC, with dedicated apps for User, Domain, Event, Task, Attendance, Announcement, Notification, Chat, and Media management.
- Developed REST APIs (DRF) for member onboarding, domain-wise team management, event lifecycle tracking, proximity-based attendance, and an automated notification/announcement pipeline.
- Built real-time group chat using **WebSockets** with support for media sharing, threaded replies, and reactions.

TECHNICAL SKILLS

Languages: Python, Go, C, C++

Backend: Django, Django REST Framework, FastAPI, Gin

AI / GenAI: LangChain, RAG, Groq API, Gemini, HuggingFace, FAISS, pgvector, Docling, Agentic workflows

Async & Background Processing: Celery, AsyncIO, Redis-queued workers, Django Channels/WebSockets

Databases: PostgreSQL, Redis

Infrastructure: Docker, Docker Compose, Nginx (TLS hardening), Linux, Git, Certbot, MinIO/S3

APIs & Auth: REST, JWT-based authentication, OTP-based auth with rate limiting

EDUCATION

Ajay Kumar Garg Engineering College, Ghaziabad, UP

2023 – 2027

B.Tech, Information Technology | **CGPA: 7.9/10**

CERTIFICATIONS

- Complete Generative AI Course with LangChain and HuggingFace** — Udemy, 62.5 hrs (Instructor: Krish Naik / KRISHAI Technologies)